

POINT LAY, AK, USA

WMO: 701210

Lat: 69.733N

Lon: 163.005W

Elev: 8

StdP: 101.23

Time Zone: -9.00 (NAK)

Period: 94-19

WBAN: 26624

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-41.0	-38.6	-41.3	0.1	-36.9	-38.5	0.1	-34.2	19.6	-20.3	17.8	-17.6	1.7	40	1.065

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.7	18.0	13.1	16.1	12.2	13.8	11.0	13.8	17.3	12.5	15.5	11.3	13.8	5.8	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.9	8.7	15.3	10.8	8.0	14.3	9.4	7.3	12.8	38.5	17.3	35.2	15.5	32.3	13.9	18.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.5	13.9	12.3	DB	-44.0	22.8	3.6	2.6	-46.6	24.7	-48.7	26.2	-50.7	27.6	-53.3	29.5	(4)
(5)			WB	-36.0	16.0	1.6	1.8	-37.1	17.3	-38.1	18.4	-39.0	19.4	-40.1	20.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-8.9	-25.1	-23.5	-23.7	-14.9	-3.0	4.9	9.5	7.2	2.0	-4.7	-14.9	-21.2	(6)
(7)		DBStd	14.11	8.02	10.03	8.64	6.93	5.74	4.04	3.58	3.10	3.42	4.53	7.73	7.96	(7)
(8)		HDD10.0	6946	1089	938	1046	748	404	161	53	97	241	454	747	967	(8)
(9)		HDD18.3	9931	1347	1171	1304	998	662	403	275	345	490	713	997	1225	(9)
(10)		CDD10.0	55	0	0	0	0	0	8	37	11	0	0	0	0	(10)
(11)		CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)
(12)		CDH23.3	3	0	0	0	0	0	2	1	0	0	0	0	0	(12)
(13)		CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)
(14)	Wind	WSAvg	5.5	5.5	5.3	5.2	5.6	6.0	5.4	5.3	5.6	5.2	5.4	4.9	5.9	(14)
(15)	Precipitation	PrecAvg	452	32	35	26	21	17	20	49	72	50	60	39	31	(15)
(16)		PrecMax	630	68	111	126	55	46	37	98	155	111	116	116	78	(16)
(17)		PrecMin	324	7	5	6	3	1	9	12	14	13	24	6	10	(17)
(18)		PrecStd	79	16	26	24	15	12	8	20	33	21	24	25	16	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-2.1	1.1	0.2	2.7	12.1	20.3	22.6	18.0	13.6	6.4	1.9	0.3	(19)
(20)			MCWB	-2.8	-0.1	-0.5	1.1	9.0	13.9	15.3	12.6	9.8	4.7	0.5	-0.3	(20)
(21)		2%	DB	-4.7	-1.4	-2.2	1.3	7.7	16.1	19.1	15.9	10.9	4.0	0.1	-2.4	(21)
(22)			MCWB	-5.3	-1.8	-2.9	0.2	5.7	11.6	13.8	12.1	8.7	3.2	-0.5	-3.1	(22)
(23)		5%	DB	-8.8	-3.7	-5.0	-2.0	5.0	13.1	17.3	13.4	8.8	2.4	-1.6	-4.7	(23)
(24)			MCWB	-9.3	-4.4	-5.7	-2.8	3.7	10.1	12.9	11.0	7.3	1.5	-2.1	-5.5	(24)
(25)		10%	DB	-13.6	-7.5	-9.8	-4.9	3.0	11.1	15.2	12.1	7.2	1.1	-3.7	-9.0	(25)
(26)			MCWB	-14.3	-8.0	-10.4	-5.7	2.0	8.5	12.0	10.2	6.0	0.4	-4.6	-9.4	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.4	0.1	-0.3	1.2	8.7	14.7	15.8	14.0	10.5	5.1	0.7	-0.1	(27)
(28)			MCDWB	-2.0	0.9	0.3	2.4	11.7	19.7	20.6	16.6	12.8	6.4	1.4	0.8	(28)
(29)		2%	WB	-4.9	-1.7	-2.6	0.1	5.7	12.2	14.4	12.4	8.5	3.1	-0.5	-2.9	(29)
(30)			MCDWB	-4.1	-1.2	-2.0	1.3	7.4	15.5	18.4	14.9	10.3	4.0	0.2	-2.4	(30)
(31)		5%	WB	-8.8	-4.0	-5.4	-2.6	3.7	10.4	13.2	11.3	7.6	1.5	-2.1	-5.2	(31)
(32)			MCDWB	-8.3	-3.2	-4.6	-2.0	5.0	13.2	16.7	13.4	8.7	2.2	-1.7	-4.4	(32)
(33)		10%	WB	-13.3	-6.9	-9.7	-5.6	2.1	8.4	12.1	10.1	6.0	0.3	-4.4	-9.5	(33)
(34)			MCDWB	-12.6	-6.3	-9.1	-4.9	3.1	10.7	15.0	11.8	7.2	0.9	-3.5	-9.0	(34)
(35)	Mean Daily Temperature Range	MDBR		7.1	7.3	7.6	7.5	5.2	6.4	6.7	5.5	4.6	5.1	6.7	6.9	(35)
(36)		5% DB	MCDBR	11.3	9.2	8.5	8.4	7.5	10.9	10.4	8.6	8.0	5.7	5.3	8.6	(36)
(37)			MCWBR	10.5	8.9	8.3	7.6	5.6	7.4	6.4	5.7	5.9	5.0	5.2	8.5	(37)
(38)		5% WB	MCDWB	10.8	9.2	8.6	8.3	7.4	10.6	9.4	8.0	7.4	5.5	5.1	8.3	(38)
(39)			MCWBR	10.4	8.9	8.4	7.5	5.7	7.4	6.0	5.4	5.7	4.9	5.0	8.2	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.163	0.216	0.247	0.311	0.355	0.337	0.346	0.341	0.304	0.233	N/A	N/A	(40)
(41)		taud		1.906	2.037	2.215	2.088	2.121	2.424	2.481	2.499	2.506	2.192	N/A	N/A	(41)
(42)		Ebn at Noon		29	528	783	811	816	857	836	796	727	505	0	0	(42)
(43)		Edh at Noon		3	41	70	112	123	95	87	76	56	34	0	0	(43)
(44)	All-Sky Solar Radiation	RadAvg		0.01	0.36	1.99	4.21	5.37	5.74	4.66	3.04	1.63	0.49	0.00	0.00	(44)
(45)		RadStd		0.00	0.04	0.11	0.22	0.48	0.44	0.55	0.32	0.15	0.06	0.00	0.00	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page